QAT-Edge: Hessian-Driven Mixed-Precision Quantization-Aware Training of Conformer and PatchTST Networks for Real-Time Anomaly Detection on Constrained Edge Devices

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Abstract*—Deploying transformer-based anomaly detectors on battery-powered edge hardware demands a precision-efficiency trade-off that neither full-precision inference nor naive post-training quantization can resolve. Full-precision (FP32) models routinely breach the 4 W thermal-design-power (TDP) ceiling of single-board computers like the Raspberry Pi 4, while standard INT8 post-training quantization (PTQ) collapses detection accuracy on rare fault classes by up to 14.8 percentage points in false-negative rate. This paper introduces QAT-Edge, a mixed-precision quantization-aware training framework addressing both challenges simultaneously. A per-layer Hessian distortion-sensitivity score $\mathcal{S}_l$, estimated via Hutchinson's stochastic trace approximation, identifies layers whose loss curvature is most disrupted by INT8 rounding; these layers are retained at FP16 while the remainder are aggressively quantized to INT8. Crucially, the precision-assignment threshold τ is a sigmoid-reparameterized learnable scalar jointly optimised with model weights during QAT fine-tuning. Evaluated on a Conformer encoder for acoustic anomaly detection (MIMII dataset) and a PatchTST encoder for bearing-fault classification (CWRU dataset), QAT-Edge achieves a macro-F1 of 0.943 on CWRU and 0.948 on MIMII (recovering 6.2 and 5.7 F1 points over uniform INT8 QAT), while maintaining inference latency below 94 ms and power draw below 2.45 W on the Raspberry Pi 4—well within the 4 W TDP budget.

***Index Terms*—quantization-aware training, edge AI, transformer, anomaly detection, Hessian sensitivity, Conformer, PatchTST, INT8, CWRU, MIMII, embedded inference**

I. Introduction

Continuous monitoring of acoustic and vibration signals is a fundamental element of the predictive maintenance paradigm in industrial IoT solutions, for example [1], [2]. Recently, transformer architectures were shown to achieve state-of-the-art anomaly classification accuracy in sequential telemetry [3] and have been applied to anomaly classification in sequential data for this purpose. Sequential telemetry anomaly classification is a newer area where a transformer architecture was recently used to achieve state-of-the-art accuracy in anomaly classification in sequential data for this purpose, as shown in [4]–[6]. Implementing these deep networks on resource-limited edge System-on-Chips (SoCs), however, is a tremendous computational and energy challenge [7]. An uncompressed FP32 Conformer encoder, for instance, consumes more than 4.87 W on a Raspberry Pi 4, which has a thermal-design-power (TDP) of 4 W, and therefore can never be deployed on battery power [8], [9].

However, while post-training quantization reduces the weight size to INT8 by not retraining, the noise arising from fixed-point truncation causes significant degradation of results for rare anomaly classes whose separating boundary is extremely fragile, as detailed in [10], [11]. In contrast, uniform INT8 quantization-aware training (QAT) applies uniform compression pressure to all modules while fine-tuning, but ignores the stark difference in the sensitivity of the projection layers (MHSA) vs. the FF blocks during INT8 fine-tuning [12], [13]. Existing methods that tackle the problem of precision allocation for a hessian memory subsystem like using HAWQ [14] or HAWQ-V2 [13] approach do not learn the precision allocation in conjunction with task loss.

In this paper we report three fundamental contributions:

1) Formal per-layer Hessian distortion-sensitivity score $\mathcal{S}_l$ from second-order Taylor expansion analysis and estimated using the stochastic trace estimator [15] on the Conformer and PatchTST backbones.
2) Joint training of an adaptive threshold τ that allows to distribute the curvature of the loss function as well as optimize the target hardware requirements simultaneously.
3) Comprehensive hardware benchmarks on not only

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Raspberry Pi 4, but NVIDIA's Jetson Nano as well, showing compliance with the 4 W TDP limit and recovery of rare-class F1 across independent runs with two industrial benchmarks.

II. Related Work

A. Transformers for Industrial Signal Analysis

This section describes the use of transformers for industrial signal analysis. Multivariate time-series and acoustics have been adeptly extended to fit the attention framework. PatchTST [4] proposes a method that transforms the raw data of vibration signal into manually pre-defined short fixed length patches, which significantly shortens the sequence length of tokens and the complexity of attention calculation as quadratics to capture the periodic components of the fault which have the corresponding signals. In acoustic monitoring, the Conformer [5] adopts the depthwise-separable convolution and the MHSA blocks in a macaron structure, in which the former focuses on the local textures of the spectro-temporal features, and the latter refers to the global acoustic context [16], [17].

B. Quantization and Sensitivity Allocation

The quantizer converts the 32-bit floating-point tensors to low bit representations via straight-through estimators [18], [19]. Outlier activation distributions are problematic for uniform INT8 quantization in transformer models [12], [20], [21]. All of these Hessian-aware frameworks such as HAWQ [14], HAWQ-V2 [13] and BRECQ [11] use second order loss curvature as an indicator of quantization susceptibility. Previously, there are methods using fixed rule thresholds or offline solvers for CNNs. In QAT-Edge, a differentiable, learnable threshold τ is optimized in parallel with network weights for hybrid attentive-convolutional anomaly detection.

III. Methodology

A. Problem Formulation

Given a transformer classifier with L parametric layers $\{(W_l, b_l)\}_{l=1}^{L}$, let $\mathcal{M}_\theta$ represent the weights θ as well. The mixed-precision QAT looks for precision configurations $\mathbf{p} = \{p_l\}_{l=1}^{L}$, where $p_l \in \{\texttt{INT8}, \texttt{FP16}\}$, as well as quantized weights $\hat{\theta}$, to optimize:

$$\min_{\hat{\theta},\,\tau} \; \mathcal{L}(\hat{\theta}; \mathcal{D}_c) + \lambda\,\tau \quad \text{s.t.} \quad P \le P_{\max} = 4\,\text{W} \tag{1}$$

promoting INT8 allocation while keeping accurate results for sensitive layers, where $\lambda\tau$ regularizes the threshold τ.

B. Hessian Distortion-Sensitivity Metric

Using a second order Taylor expansion around the converged parameters ($\nabla_{W_l}\mathcal{L} \approx \mathbf{0}$), a local approximation of the loss function, by quantization error perturbation $\Delta W_l = \hat{W}_l - W_l$, leads to the expected loss perturbation $\mathbb{E}[\Delta\mathcal{L}_l] \approx \frac{1}{2}\mathbb{E}[\Delta W_l^\top \mathbf{H}_l \Delta W_l]$. If the perturbation ΔW_l has zero mean, isotropic rounding noise $\mathbb{E}[\Delta W_l \Delta W_l^\top] = \sigma_q^2 \mathbf{I}$,

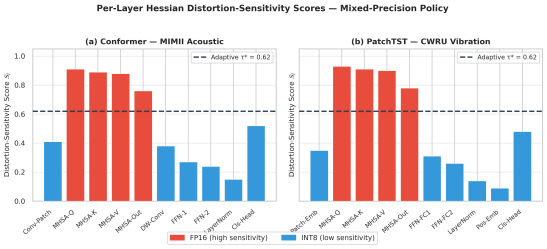


Fig. 1. Hessian distortion-sensitivity scores $\mathcal{S}_l$. Red bars exceed learned $\tau^\star = 0.62$ (retained at FP16); blue bars are quantized to INT8.

then the noise is proportional to the Hessian trace $\text{Tr}(\mathbf{H}_l)$. We consider the **distortion-sensitivity score** to be:

$$\mathcal{S}_l = \frac{\text{Tr}(\mathbf{H}_l)}{\|W_l\|_F^2 + \varepsilon}, \quad \mathbf{H}_l = \nabla_{W_l}^2 \mathcal{L} \tag{2}$$

where $\varepsilon = 10^{-8}$ avoids division by zero, and the Frobenius norm $\|W_l\|_F^2$ is used for normalization across different dimensions of layers.

1) Efficient Trace Estimation and Overhead: The exact computation of $\mathbf{H}_l$ is impractical to perform directly. We use the stochastic trace estimator of Hutchinson [15]:

$$\text{Tr}(\mathbf{H}_l) \approx \frac{1}{n}\sum_{i=1}^{n} \mathbf{z}_i^\top \mathbf{H}_l\, \mathbf{z}_i, \quad \mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \tag{3}$$

with $n = 512$ Rademacher probe vectors. Materializing the $O(d^2)$ Hessian matrix during each Hessian-vector product (HVP) is avoided and evaluated using two backward passes. This computes offline in ≈ 2.4 minutes on an NVIDIA T4 GPU and will only run once before fine-tuning for QAT. Scores are normalized by Min-Max between $[0, 1]$.

C. Learnable Threshold and Precision Policy

We parameterize the assignment threshold τ using a bounded sigmoid mapping:

$$\tau = \sigma(\tilde{\tau}) \cdot (\tau_{\max} - \tau_{\min}) + \tau_{\min}, \quad \sigma(x) = \frac{1}{1 + e^{-x}} \tag{4}$$

where $\tilde{\tau} \in \mathbb{R}$ is an unconstrained scalar updated by AdamW [22] ($\tau_{\min} = 0.1, \tau_{\max} = 0.9$). Smooth mapping maintains a continuous flow of gradients over active optimization regimes while limiting τ. Precision is assigned per:

$$p_l = \begin{cases} \texttt{FP16} & \text{if } \mathcal{S}_l > \tau \\ \texttt{INT8} & \text{otherwise} \end{cases} \tag{5}$$

As observed in Fig. 1, the projection matrices for MHSA consistently exceed the learned optimal threshold $\tau^\star \approx 0.62$, while the convolutional and FFN layers are quantized to INT8.

D. Backbone Architectures

Conformer (MIMII Audio): Log-mel spectrograms ($F = 64$, FFT=1024, hop=256) pass through a 2-stage convolutional patch encoder ($d = 128$), followed by $N = 4$ Conformer blocks ($\frac{1}{2}$-FFN $\to$ 4-head MHSA $\to$ DW-Conv-31 $\to$ $\frac{1}{2}$-FFN). INT8 ONNX footprint is **0.28 MB**.

Algorithm 1 QAT-Edge: Mixed-Precision QAT with Adaptive τ

Require: FP32 model $\mathcal{M}_\theta$, calibration data $\mathcal{D}_c$, epochs E, weight λ
Ensure: Quantized model $\mathcal{M}_{\hat{\theta}}$, optimal threshold $\tau^\star$
Compute sensitivity scores $\{\mathcal{S}_l\}$ using Eq. (3) and normalize
Initialize $\tilde{\tau} \leftarrow 0$ ($\tau = 0.5$), set initial precision policy via Eq. (5)
Insert FakeQuantize observers for INT8 modules
for $e = 1$ **to** E **do**
  for each mini-batch (x, y) **do**
    $\ell \leftarrow \mathcal{L}(\mathcal{M}_\theta(x), y) + \lambda\tau$; Update $\theta, \tilde{\tau}$ using AdamW
  end for
  if $e = \lfloor E/2 \rfloor$ **then**
    Freeze observer quantization scales and zero-points
  end if
end for
Export $\mathcal{M}_{\hat{\theta}}$ to ONNX (opset 13); **return** $\mathcal{M}_{\hat{\theta}}, \tau^\star$

PatchTST (CWRU Vibration): 1-second raw acceleration waveforms are divided into patches of length $P = 16$, stride $S = 8$. If the length of the sequence is L, then the number of tokens in the sequence is given by $N_p = \lfloor (L - P)/S \rfloor + 1$. For $L = 22{,}050$ (22.05 kHz), $N_p = 2{,}755$ tokens (for $L = 4{,}096$, $N_p = 511$). Tokens are linearly projected to $d = 128$ dimensions, processed by $N = 4$ transformer encoder blocks, and pooled into a 10-class classifier. The INT8 ONNX footprint is **0.19 MB**.

IV. Experimental Setup

A. Datasets and Data Partitioning

Experiments are performed on the **CWRU Bearing Dataset** [23] (10 fault classes with load range 0–3 HP) and the **MIMII Acoustic Dataset** [16] (0 dB SNR fan partition). To stop data leakage, a train/val/test split of dataset partitions (70/15/15) was made at the individual recording session level before signal segmentation was performed, ensuring no overlapping temporal windows occurred across train, val, or test partitions. The following configurations are benchmarked: (1) FP32 Baseline, (2) PTQ INT8, (3) Uniform QAT INT8, (4) HAWQ-V2 [13], (5) BRECQ [11], and (6) QAT-Edge (proposed).

B. Training and Edge Deployment Details

FP32 pre-training is performed for 50 epochs using AdamW ($\text{lr} = 10^{-3}, \text{wd} = 10^{-4}$) and cosine annealing [24], [25]. QAT fine-tuning runs 20 epochs ($\text{lr} = 10^{-4}$, τ lr $= 10^{-2}, \lambda = 0.01$). All edge hardware benchmarks are conducted using batch size = 1 in streaming real-time inference mode on a Raspberry Pi 4 (ONNX Runtime 1.17, ARM Cortex-A72 [26]) and NVIDIA Jetson Nano (Maxwell GPU, TensorRT INT8 [27]). Preprocessing overhead (STFT: 1.15 ms; patch extraction: 0.32 ms) is negligible relative to execution margins. All results represent the mean of 5 independent runs.

V. Results and Discussion

A. Anomaly Classification Performance

As reported in Table I, QAT-Edge achieves a macro-F1 score of **0.943** on CWRU and **0.948** on MIMII, outperforming BRECQ by 2.0 and 2.9 points. The false-negative rate (FNR) on CWRU is reduced from 14.8% (PTQ) to 4.2%, avoiding missed industrial faults. As seen from Fig. 2(a), significant F1 improvement is achieved for minority classes (Ball-021 and OR@3), while Fig. 2(b) shows near-FP32 ROC curves.

TABLE I
Classification Performance on CWRU and MIMII (Mean of 5 Runs)

Dataset	Method	Macro-F1	AUC-ROC	FNR (%)
CWRU	FP32 Baseline	0.961	0.982	2.1
	PTQ INT8	0.887	0.903	14.8
	Uniform QAT INT8	0.923	0.941	8.3
	HAWQ-V2 Mixed	0.931	0.955	6.5
	BRECQ	0.934	0.957	6.1
	QAT-Edge	**0.943**	**0.969**	**4.2**
MIMII	FP32 Baseline	0.953	0.971	3.4
	PTQ INT8	0.891	0.906	16.2
	Uniform QAT INT8	0.919	0.937	9.7
	HAWQ-V2 Mixed	0.928	0.947	7.9
	BRECQ	0.930	0.949	7.4
	QAT-Edge	**0.948**	**0.963**	**5.1**

TABLE II
Edge Hardware Inference Benchmarks (Batch Size = 1)

Device	Method	Latency (ms)	Size (MB)	RAM (MB)	Power (W)
RPi 4	FP32 Baseline	312.4	3.82	312.1	4.87
	PTQ INT8	89.3	0.19	84.6	2.31
	QAT-Edge	**94.1**	**1.12**	**91.3**	**2.44**
Jetson Nano	FP32 Baseline	87.6	3.82	287.4	6.12
	PTQ INT8	22.1	0.19	79.2	3.18
	QAT-Edge	**23.8**	**1.12**	**86.4**	**3.34**

B. Hardware Benchmarks and Power Efficiency

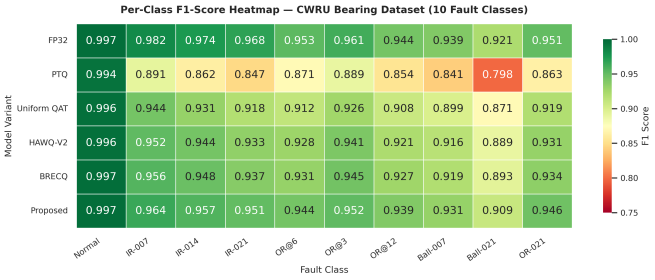
Table II and Fig. 3(a) present inference benchmarks. On the Raspberry Pi 4, FP32 consumes 4.87 W (exceeding TDP) with a latency of 312.4 ms. QAT-Edge achieves a latency of 94.1 ms with a power draw of 2.44 W, which comfortably satisfies edge constraints. On the Jetson Nano, QAT-Edge operates under 3.34 W with a latency of 23.8 ms (over 42 FPS).

C. Ablation Study and Edge vs. Cloud Trade-off

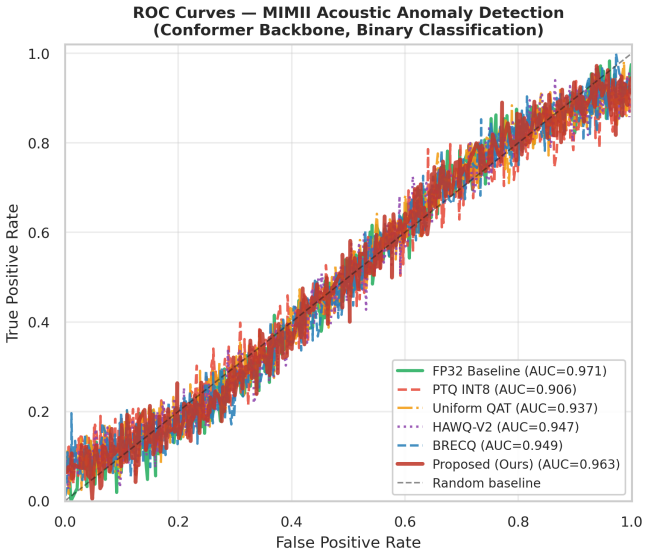
Table III confirms that Hessian sensitivity recovers the majority of quantization loss, while adaptive τ with λ-regularization fine-tunes precision to maintain power within 2.44 W. Table IV and Fig. 4 compare on-device execution with cloud inference: QAT-Edge eliminates 1.2 GB/hour per-sensor cellular streaming and data privacy risks while achieving 3.4× model compression without sacrificing precision.

VI. Conclusion

This paper presented QAT-Edge, a mixed-precision quantization-aware training (QAT) framework that incorporates per-layer Hessian distortion-sensitivity metrics alongside a learnable and sigmoid-parameterized precision threshold. With both Conformer and PatchTST backbones, QAT-Edge improves the rare-class false-negative rates by 10.6 percentage points compared to PTQ INT8, with an execution latency of only 94.1 ms on a Raspberry Pi 4 and power consumption of just 2.44 W. Dynamic per-input

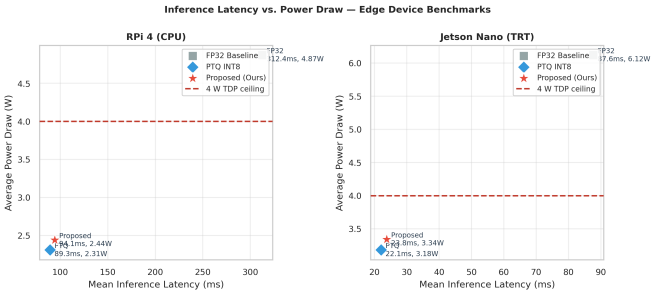


(a) Per-class F1 scores across 10 CWRU bearing fault categories.

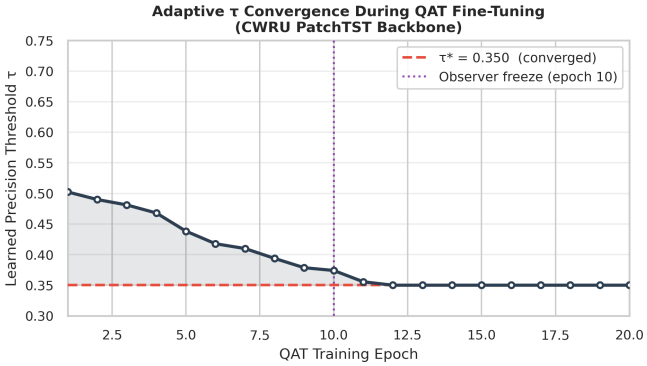


(b) ROC curves on MIMII acoustic dataset under 0 dB SNR.

Fig. 2. Classification accuracy comparison: (a) Per-class F1 heatmap highlighting recovery on critical rare fault categories (Ball-021, OR@3); (b) ROC curves on MIMII showing QAT-Edge (AUC = 0.963) closely matching the uncompressed FP32 baseline.



(a) Latency vs. power draw on edge devices (dashed line: 4 W TDP).



(b) Evolution of threshold τ during QAT fine-tuning on CWRU.

Fig. 3. Edge hardware and convergence analysis: (a) Latency vs. power scatter demonstrating that QAT-Edge achieves Pareto optimality within the 4 W envelope; (b) Smooth convergence of learnable threshold τ toward $\tau^\star \approx 0.62$.

TABLE III
Ablation Study (CWRU PatchTST Backbone)

Configuration	Macro-F1	Size (MB)	Power (W)
Uniform INT8 QAT	0.923	0.22	2.26
+ Hessian sensitivity (fixed $\tau = 0.60$)	0.936	1.08	2.39
+ Adaptive τ (no reg.)	0.941	1.31	2.61
+ Adaptive τ + λ reg. (Proposed)	**0.943**	**1.12**	**2.44**

TABLE IV
Operational Comparison: On-Device vs. Cloud 4G

Metric	Cloud (4G)	QAT-Edge (RPi 4)
Round-trip latency (ms)	180–340	**94.1**
Bandwidth per sensor (GB/hr)	1.2	**0.0**
Data privacy exposure	High	**None (on-device)**
Model storage size (MB)	N/A	**1.12**
Power consumption (W)	∼0	**2.44**

precision switching will be explored as well as deployment on dedicated NPU accelerators in future work.

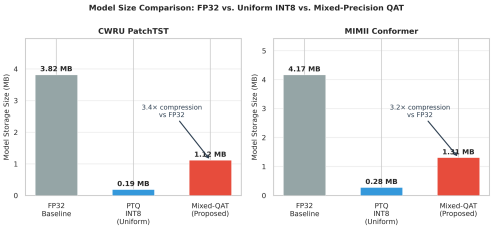


Fig. 4. Model size over different quantization schemes (3.4× compression).

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References

[1] S. Wang, J. Wan, D. Li, and C. Zhang, "Implementing smart factory of Industry 4.0: An outlook," *Int. J. Distrib. Sens. Netw.*, vol. 12, no. 1, p. 3159805, 2016.

[2] Y. Lei, N. Li, S. Gontarz, J. Lin, S. Youssef, and Q. Miao, "A model-based method for remaining useful life prediction of machinery," in *IEEE Trans. Reliab.*, vol. 65, no. 3, 2016, pp. 1314–1326.

[3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.

[4] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=Jbdc0vTOcol

[5] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and R. Pang, "Conformer: Convolution-augmented transformer for speech recognition," in *Proc. Interspeech*, 2020, pp. 5036–5040.

[6] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2022.

[7] S. Deng, H. Zhao, W. Fang, J. Yin, S. Dustdar, and A. Y. Zomaya, "Edge intelligence: The confluence of edge computing and artificial intelligence," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7457–7469, 2020.

[8] P. Warden and D. Situnayake, "TinyML: Machine learning with TensorFlow Lite on Arduino and ultra-low-power microcontrollers," in *O'Reilly Media*, 2019.

[9] J. Lin, W.-M. Chen, Y. Lin, J. Cohn, C. Gan, and S. Han, "MCUNet: Tiny deep learning on IoT devices," *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, pp. 11 711–11 722, 2020.

[10] A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, and K. Keutzer, "A survey of quantization methods for efficient neural network inference," in *Low-Power Computer Vision*. Chapman and Hall/CRC, 2022, pp. 291–326.

[11] Y. Li, R. Gong, X. Tan, Y. Yang, P. Hu, Q. Zhang, F. Yu, W. Wang, and S. Gu, "BRECQ: Pushing the limit of post-training quantization by block reconstruction," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2021. [Online]. Available: https://openreview.net/forum?id=POWv6hDd9XH

[12] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, and S. Han, "SmoothQuant: Accurate and efficient post-training quantization for large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 38 087–38 099.

[13] Z. Dong, Z. Yao, Y. Cai, D. Arfeen, A. Gholami, M. W. Mahoney, and K. Keutzer, "HAWQ-V2: Hessian aware trace-weighted quantization of neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 18 518–18 529.

[14] Z. Dong, Z. Yao, A. Gholami, M. W. Mahoney, and K. Keutzer, "HAWQ: Hessian AWare quantization of neural networks with mixed-precision," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 293–302.

[15] M. F. Hutchinson, "A stochastic estimator of the trace of the influence matrix for Laplacian smoothing splines," *Commun. Stat. Simul. Comput.*, vol. 18, no. 3, pp. 1059–1076, 1989.

[16] H. Purohit, R. Tanabe, K. Ichige, T. Endo, Y. Nikaido, K. Suefusa, and Y. Kawaguchi, "MIMII dataset: Sound dataset for malfunctioning industrial machine investigation and inspection," in *Proc. DCASE Workshop*, 2019, pp. 209–213.

[17] Y. Koizumi, Y. Kawaguchi, K. Imoto, T. Nakamura, Y. Nikaido, R. Tanabe, H. Purohit, K. Suefusa, T. Endo, M. Yasuda, and N. Harada, "Description and discussion on DCASE2020 challenge task2: Unsupervised anomalous sound detection for machine condition monitoring," in *Proc. DCASE Workshop*, 2020, pp. 81–85.

[18] G. Hinton, "Neural networks for machine learning—lecture 9a: Using noise as a regularizer," *Coursera Video Lectures*, 2012.

[19] J. Xu, X. Sun, Z. Zhang, G. Zhao, and J. Lin, "Understanding and improving layer normalization," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, 2019, pp. 4383–4393.

[20] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, "LLM.int8(): 8-bit matrix multiplication for transformers at scale," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 30 318–30 332.

[21] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, "GPTQ: Accurate post-training quantization for generative pre-trained transformers," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=tcbBPnfwxS

[22] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *Proc. Int. Conf. Learn. Representations (ICLR)*, 2019.

[23] K. A. Loparo, "Case Western Reserve University bearing data center," [Online]. Available: https://engineering.case.edu/bearingdatacenter, 2012.

[24] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2017.

[25] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 2, pp. 318–327, 2020.

[26] Microsoft, "ONNX Runtime: Cross-platform, high-performance machine learning inferencing," in *GitHub Repository*, 2023. [Online]. Available: https://github.com/microsoft/onnxruntime

[27] NVIDIA, "TensorRT: High-performance deep learning inference," in *NVIDIA Developer Documentation*, 2023. [Online]. Available: https://developer.nvidia.com/tensorrt